OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TECHNOLOGICAL UNIVERSITY DUBLIN

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BLANCHARDSTOWN

Academic Term	2022-23
Year of Study	Year 2
Semester	Semester 2 – Autumn
Date of Examination	
Time of Examination	

Programme Code	Programme Title	Module Code
TU653	Higher Certificate in Science in Computing in information Technology	COMP H2014
TU757	Bachelor of Science in Computing in Information Technology	COMP H2014
TU860	Bachelor of Science (Honours) in Computing	COMP H2014
TU758	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2014
TU863	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2014

SEMESTER TWO AUTUMN EXAM

Module Title	Operating Systems

Internal Examiner	**Dr. Kevin Farrell**
External Examiner	**Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Question 1 in Section A is COMPULSORY. Candidates should attempt ALL parts of Question 1 in Section A, and ANY <u>two</u> other questions in Section B.**
3.	**Where a question asks you to use diagrams, to obtain full marks, you MUST provide diagrams.**
4.	**Where a question asks you to use code, to obtain full marks, you MUST provide code.**
5.	**This paper is worth 100 marks. Question 1 is worth 40 marks. All other questions are worth 30 marks each.**
6.	**Please make sure you write the question numbers you answer in the boxes on the front of your answer book.**
7.	**This is a two hour examination.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A: COMPULSORY QUESTION

Question 1: Answer *ALL* parts of this question *(4 marks each part)*
Please write the number of this question on the front of your answer book

a) In relation to Linux, explain what a *Makefile* is. List and briefly describe its **three** essential elements.

b) In multi-core programming, distinguish between *data parallelism* and *task parallelism*.

c) In relation to a process, explain the distinction between *instructions* and *data*, and the relationship between them.

d) Briefly describe **four** items contained in a *Process Control Block*.

e) Explain the mechanism for a process to switch from *user mode* to *kernel mode* in an Operating System.

f) Give your reasoned opinion as to why it is considered bad practice to log directly into a Linux GUI-based system as *root*, as opposed to logging into the GUI as an ordinary user, and then using a terminal window to become *root*?

g) Describe the *modular kernel methodology* of Operating Systems design.

h) Explain the following terms with regard to Process Management: *pre-emptive scheduling, context change, multiprogramming, program counter*.

i) Describe the *Bounded-Buffer Producer-Consumer Problem*, detailing why a synchronisation solution is necessary.

j) Describe the **four** simultaneous conditions necessary in order for *deadlock* to occur.

(40 Marks Total)

Please now write the number of this question in the box on the front of your answer book.

SECTION B: ANSWER ANY *TWO* QUESTIONS

Question 2

If you answer this question please write the number on the front of your answer book

a) Consider the situation where a user downloads the source-code *tarball* of an application they wish to install on their Linux system. Using the **`tar`** command, the user has extracted the files from the *tarball* into their home directory, has changed into the program directory created, and now wishes to install the application. List all of the commands that the user must type to *configure*, *compile* and *install* the software on their system. For each step, provide the command prompt symbol to indicate whether an *ordinary user* or *root* should type the command.

(5 marks)

b) Consider the following situation:

- A user creates a script file, called *myscript* in their home directory of a Linux system.
- They try to run this script by changing to their home directory, and typing the script name at the $ prompt in a terminal window, as follows:

```
$ myscript
```

- But, the shell returns the error *"bash: myscript: command not found"*.
- They have correctly debugged the script, and have also set the correct executable permissions on the file.

What do they need to type in order to run it. Give a reason for your answer.

(3 marks)

This Question is continued on the next page…

c) Describe, with the aid of a diagram, the layout of the Linux Filesystem. Your answer must include **six** directories *directly* under the *root* directory, with an explanation of the purpose of each directory.

(8 marks)

d) Describe the UNIX/Linux *File Permissions* model. Your answer should include an explanation, with an example, of how the permissions on a file or directory can be represented as a set of *octal* integers.

(8 marks)

e) Explain, with the aid of a diagram, how a process is created on UNIX/Linux systems when a user types a command in the *foreground* in the console.

(6 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.

Question 3

If you answer this question please write the number on the front of your answer book

a) Explain the concept of *multi-threaded processes*, and how they differ from single-threaded processes. Use diagram(s) to aid your answer.

(5 marks)

b) Describe any one of the **four** common models of *threading* implementation in modern computing systems.

(6 marks)

c) Describe the operating systems issues associated with multi-threaded processes under any **three** of the following headings:

- Semantics of fork() and exec() system calls
- Thread Cancellation
- Signal Handling
- Thread Pools
- Thread Specific Data

(15 marks)

d) In a particular program, the serial fraction of code is **5%** (i.e. **5%** of the code is not parallelisable). According to *Amdahl's Law* (see below), what is the speed-up achievable for this program, when run on a multi-core system with **32** cores?

Amdahl's Law:

$$\text{Speed-Up} \leqslant \frac{1}{S + \frac{(1-S)}{N}}$$

where:

- S = Serial Fraction of Code
- N = Number of CPU Cores

(4 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.

Question 4

If you answer this question please write the number on the front of your answer book

a) In relation to process synchronisation, provide the **three** conditions that need to be satisfied in order to solve the *critical-section problem.*

(6 marks)

b) Describe the *First Readers-Writers Problem* of process synchronisation. Explain why a synchronisation solution is necessary, and outline any problems such a solution may have.

(7 marks)

c) Explain why the traditional (*Dijkstra*) definition (given below) of the atomic operation *wait(S)* on the semaphore, *S*, is often referred to as a *spinlock*. To aid your explanation, provide a general program pseudo-code solving the critical section problem using semaphores.

```
wait(S)
{
     while(S <= 0)
           ; // Do nothing
     S--;
}
```

(9 marks)

d) Explain, with the aid of diagrams indicating the execution of sample processes, the circumstances where the use of *spinlocks* (referenced in **part c)** above) is advantageous.

(8 marks)

(30 Marks Total)

Did you answer this question? If so, please now write the number of this question in the box on the front of your answer book.